

Improving Cultural Knowledge Coverage in LLMs via Region-Aware Retrieval-Augmented Generation: An Extension of BLEnD

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I. PROJECT DETAILS

Criterion followed: Code, dataset, and paper are publicly available (Myung et al., 2024, NeurIPS Datasets & Benchmarks Track [1]) – “If code available, address limitation and proposed solution.” We identify a concrete limitation of the released baseline (large, uneven performance gap between high-resource and low-resource cultures, see Section II) and propose and implement three inference-time interventions – region-aware RAG, an adaptive retrieval-need gate, and cross-lingual retrieval pivoting – to reduce that gap, using only the officially released dataset and evaluation code. (The checklist’s “dataset not available” branch is therefore N/A: BLEnD’s data, code, and evaluation script are all publicly released, as cited in Section IV.)

Problem-statement (PS) decomposition. We decompose the project into four problem statements, each a self-contained, independently-verifiable slice of the pipeline, and split them evenly, two per member:

- **PS-1 – Baseline Replication & Region-Aware RAG** (foundational, Member 1): exact replication of BLEnD’s zero-shot + persona baseline, and the region-aware hybrid-retrieval RAG condition.
- **PS-2 – Adaptive Retrieval Decision Layer** (Member 2): a lightweight retrieval-need gate that decides, per question, whether RAG should fire at all.
- **PS-3 – Cross-Lingual Retrieval** (Member 1): pivoting low-resource-language questions into an English-compatible retrieval representation against an English cultural knowledge base.
- **PS-4 – Category-Aware Analysis** (Member 2): systematic Country $\times$ Category and Method $\times$ Category analysis of where retrieval helps, hurts, or is neutral.

Fig. 1 shows the ownership map. Both members jointly own the shared experiment matrix, the official BLEnD `evaluate.sh` scoring, and the final report.

Team Composition and Individual Contributions:

- **Member 1:** Gaurav Kumar, 12340790, Email ID: gauravkr@iitbhillai.ac.in
Contribution (PS-1 + PS-3 – Data, Baseline & Cross-Lingual Track): BLEnD dataset subsetting and pre-processing for the 6 selected countries; implementation of the zero-shot + persona baseline prompting pipeline (replicating the original paper’s method) on Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, and Meta Llama 3.1; running and logging all baseline inference calls;

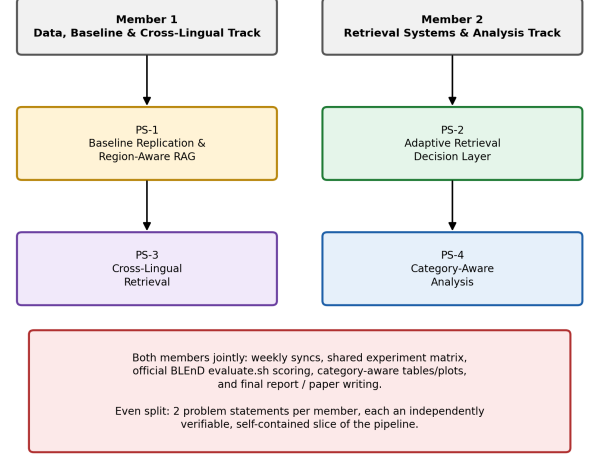


Fig. 1. Problem-statement ownership map: four problem statements (PS-1–PS-4) split evenly, two per member, along the same non-adaptive/adaptive-and-analysis axis as the project’s original two-track design.

construction of the per-country cultural knowledge base (non-BLEnD sources, to avoid contamination) and the hybrid BM25 + dense-embedding retrieval module used by the RAG condition; design and implementation of the cross-lingual pivot (Section III-C) and the local-vs-English-pivot retrieval comparison on the low-resource-language countries; adapting and running the official `evaluate.sh` scoring pipeline; producing baseline and cross-lingual result tables.

- **Member 2:** Siddharth Rai, 12342050, Email ID: siddharthp@iitbhillai.ac.in

Contribution (PS-2 + PS-4 – Retrieval Systems & Analysis Track): implementation of the RAG-augmented generation pipeline on Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, and Meta Llama 3.1; running all Always-RAG-condition inference calls; design and implementation of the adaptive retrieval-need decision layer (Section III-B) and the Adaptive-RAG condition; running all Adaptive-RAG inference calls; ablation analysis (Baseline vs. RAG vs. Adaptive-RAG, per country, per model) and error analysis; category-aware analysis (Section III-D) – Country $\times$ Category and Method $\times$ Category tables/plots – computed from the shared result logs rather than additional LLM calls.

Both members jointly participate in weekly plan reviews, results discussion, and the final report/paper writing, ensuring equal ownership of the project’s outcome. The PS-1/PS-3 vs. PS-2/PS-4 split keeps the same design principle as the original two-track split: each member owns one full, comparable, independently-producible half of the pipeline (the non-adaptive data/baseline/cross-lingual side vs. the adaptive-retrieval/analysis side), avoiding a dependency where one member blocks the other’s numbers. Category-aware analysis (PS-4) deliberately reuses outputs already produced by PS-1/PS-2/PS-3 rather than issuing new LLM calls, so it does not expand the project’s API budget.

II. PROBLEM STATEMENT

Context and motivation. Large Language Models (LLMs) are trained predominantly on English-centric, Western-skewed web data. As a result, they perform well on everyday cultural-knowledge questions for highly-represented regions (e.g., the US), but perform poorly for underrepresented cultures and low-resource languages, even for mundane, non-controversial facts such as typical foods, school subjects, or holiday customs. BLEnD [1] is a NeurIPS 2024 benchmark that quantifies this gap directly: it contains 52.6k hand-annotated question-answer pairs across 16 countries/regions and 13 languages, and shows that state-of-the-art LLMs (e.g., GPT-4) score 79.22% on US culture in English but only 12.18% on Ethiopian culture in Amharic – a 67-point gap on identically-structured questions.

Why this matters. As LLMs are deployed globally, this imbalance translates into unequal quality of service, reinforced stereotyping of underrepresented cultures, and reduced usefulness for the majority of the world’s population who are not from high-resource, Western, English-speaking contexts. This is an active, recognised research problem: BLEnD was subsequently extended into **SemEval-2026 Task 7** [2], a shared task (33 language-culture pairs, 140+ registered participating teams) run specifically to benchmark systems that try to close this gap using inference-time methods, since the task explicitly disallows fine-tuning on the benchmark itself.

Specific scope of our work. Reproducing the full BLEnD evaluation (16-24 models, GPU clusters, multiple paid APIs) is outside the resource envelope of a 2-person, 2-month, zero-budget project. We therefore scope our contribution as a broader question than a single baseline-vs-RAG comparison: *can lightweight, inference-time retrieval interventions – region-aware RAG, an adaptive retrieval-need gate, and cross-lingual retrieval pivoting – reduce the accuracy gap between high-resource and low-resource cultures on BLEnD’s Short-Answer Question (SAQ) track, without any model fine-tuning and using only freely-available LLM access, and do these effects vary systematically across BLEnD’s six knowledge categories?* We test this on 6 countries spanning the high-to-low resource spectrum (United States, Spain, South Korea, Azerbaijan, Assam, Ethiopia) using five free, non-proprietary, locally deployable open-weight LLMs (Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, Meta Llama 3.1). Concretely, we structure the work as four problem statements (PS-1–PS-4,

detailed in Section I and Section III): (1) baseline replication and region-aware RAG, (2) an adaptive retrieval decision layer (Baseline vs. Always-RAG vs. Adaptive-RAG), (3) cross-lingual retrieval for low-resource-language countries, and (4) category-aware analysis of where retrieval helps, hurts, or is neutral. Our primary reported result remains the per-country, per-model, per-condition accuracy delta, now extended with a category-level breakdown.

III. METHODOLOGY

A. PS-1: Baseline Replication & Adaptive Hybrid RAG (Member 1 / Member 2, foundational)

Baseline (replicated from the original paper). For each of the 6 selected countries, we replicate BLEnD’s own evaluation protocol: each question is answered under two prompt settings – (1) direct instruction, and (2) persona prompting (“You are a person from {country}...”), averaged per the paper’s method – using zero-shot inference, no retrieval, no fine-tuning.

Proposed method: Adaptive Hybrid RAG. We augment the same prompting protocol with a region-aware hybrid retrieval step, followed by a lightweight adaptive gate in PS-2:

- 1) **Knowledge base construction:** For each of the 6 target countries, we curate a small corpus of cultural-context passages from public non-BLEnD sources (Wikipedia country/culture pages, Wikivoyage). This is a deliberate design choice: the SemEval-2026 follow-up competition on this exact benchmark found that some LLMs’ apparent gains on BLEnD were inflated by training-data contamination (models had likely seen BLEnD’s own 2024 public release) [2]. Sourcing our KB independently avoids this pitfall.
- 2) **Hybrid retrieval:** For each question, we retrieve the top- k passages using a combination of BM25 (lexical) and dense embedding similarity (semantic), restricted to the target country’s documents – a design inspired by the region-aware hybrid retrieval strategy used by Team Simorgh [3] in the SemEval-2026 shared task on this benchmark.
- 3) **Grounded generation:** Retrieved passages are concatenated with the original persona-conditioned prompt and passed to the selected open-weight LLM to generate the final short answer.

Key technique – selective evaluation of RAG’s effect. Prior work on this exact task (Team CultRAG, SemEval-2026 [4]) found that RAG helps underrepresented cultures but can *hurt* performance on already-high-resource, high-consensus cultures due to retrieval noise. Rather than assuming a uniform improvement, our central experiment is a controlled **per-country ablation**: baseline vs. RAG accuracy, separately for each of the 6 countries and each of the 5 models, to test directly whether the gap between high- and low-resource cultures narrows.

How this differs from existing/reproduced work. The BLEnD paper itself only evaluates zero-shot prompting (no retrieval). The best-performing system in the related SemEval-2026 shared task (Team king001 [5]) also used RAG, but its reported gains were confounded by contamination and its final

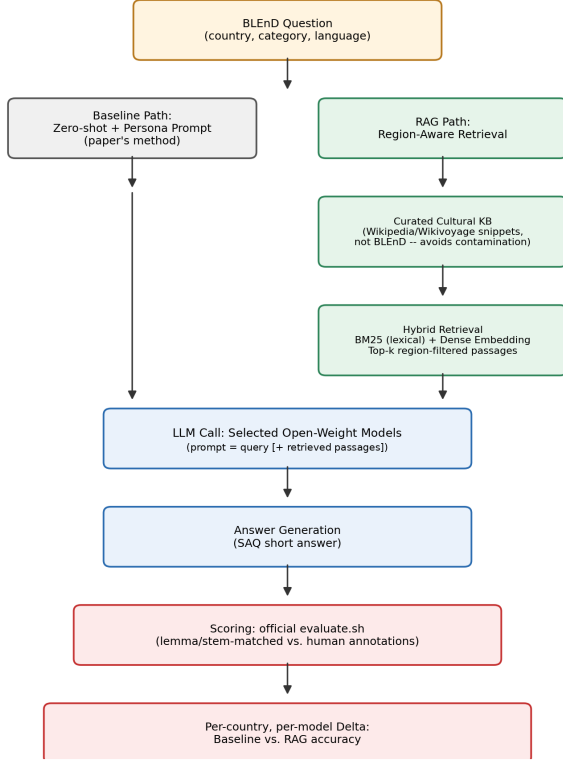


Fig. 2. PS-1 pipeline: baseline (zero-shot + persona prompting, replicating the original paper) is run alongside a region-aware RAG path (independent knowledge base → hybrid BM25 + dense retrieval → grounded generation). Both paths are scored with the official BLENd `evaluate.sh` script, and compared per country and per model.

margin over a plain baseline was effectively zero on uncontaminated held-out data. Our contribution is: (a) a contamination-safe knowledge base built independently of BLENd, (b) an ablation explicitly designed to detect *where* retrieval helps or hurts rather than reporting a single aggregate number, and (c) full reproducibility using free, locally deployable resources only.

Figure 2 shows the end-to-end baseline-vs-RAG pipeline.

B. PS-2: Adaptive Retrieval Decision Layer (Member 2)

Motivation. Retrieval is not always beneficial: an irrelevant or noisy passage can distract the LLM even when the model already “knows” the answer, which is exactly the failure mode Team CultRAG reported for high-resource cultures [4], and echoes the broader adaptive-retrieval finding that indiscriminately retrieving passages for every query can hurt versatility and answer quality [6]. We therefore extend the baseline-vs-RAG comparison to a three-way comparison: **Baseline vs. Always-RAG vs. Adaptive-RAG**.

Definition (project-specific, deliberately lightweight). “Adaptive retrieval” here means a per-question, training-free *gate* that decides whether to invoke the region-aware RAG path at all, rather than a learned or fine-tuned router. We do not fine-tune any component; the gate is a heuristic computed at inference time from signals already available in the pipeline:

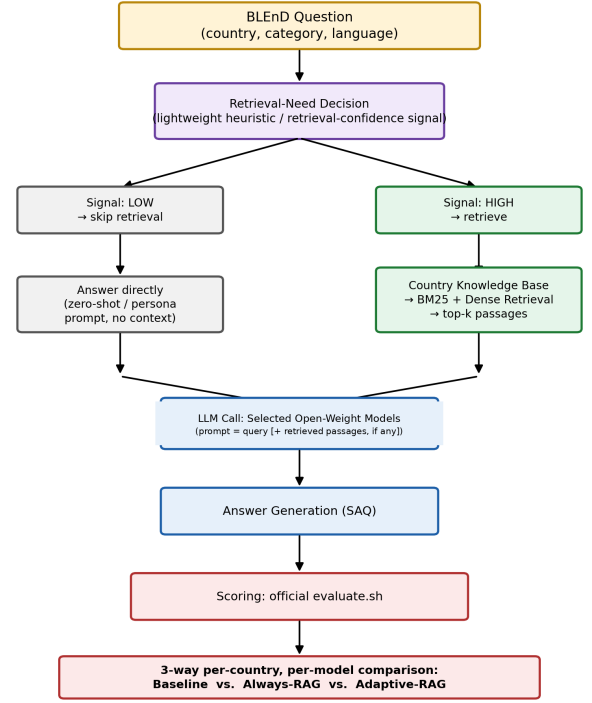


Fig. 3. PS-2 pipeline: a lightweight, training-free retrieval-need gate decides per question whether to skip retrieval (answer directly) or invoke the region-aware RAG path from Fig. 2, producing a three-way Baseline vs. Always-RAG vs. Adaptive-RAG comparison.

- **Retrieval-confidence signal:** the top- k hybrid-retrieval score (BM25 + dense similarity) for the best-matching passage. If the top passage’s similarity falls below a threshold tuned on a small held-out slice, we treat retrieval as unlikely to help and skip it.
- **Question-type signal (cheap heuristic):** category and country resource-level (from BLENd’s own Joshi et al. resource-class taxonomy) as a coarse prior – e.g., high-resource-country/common-category questions are more likely to be already “known” by the LLM.

Either signal alone is enough to implement within our constraints; we will use whichever proves more predictive on a small pilot (Week 3), and report which one we used and why.

Experimental conditions. (1) *Baseline* – no retrieval, ever. (2) *Always-RAG* – retrieve for every question (PS-1). (3) *Adaptive-RAG* – retrieve only when the gate fires. Fig. 3 shows the decision flow.

Why Adaptive-RAG could outperform Always-RAG. If retrieval only helps on genuinely knowledge-sparse questions and adds noise elsewhere, Adaptive-RAG should match Always-RAG’s gains on low-resource cultures while avoiding some of Always-RAG’s losses on high-resource cultures – i.e., a better accuracy/noise trade-off, not necessarily a higher ceiling. We do *not* claim Adaptive-RAG will beat Always-RAG outright; this is the hypothesis under test, not an assumed result.

Ablations. Per-country, per-model comparison of all three conditions; fraction of questions for which the gate chooses to retrieve, broken down by country resource-level; and a

confusion-style analysis of cases where the gate’s choice (retrieve/skip) was “wrong” in hindsight (i.e., Always-RAG beat Baseline but the gate skipped, or vice versa).

Implementation cost. No fine-tuning, no additional model calls beyond the retrieval step already computed for PS-1 – the gate reuses the retrieval score that hybrid retrieval already produces, so Adaptive-RAG adds negligible extra compute over Always-RAG.

C. PS-3: Cross-Lingual Retrieval (Member 1)

Motivation. BLEnD’s own finding is that for *low-resource* languages, LLMs answer better in English than in the local language, while the reverse holds for high-resource languages [1]. This suggests that for our three low-resource countries (Azerbaijan, Assam, Ethiopia), a local-language question might retrieve more useful evidence if it is first pivoted into an English-compatible representation and matched against an *English* cultural knowledge base, rather than searched directly in the local language against a (likely sparser) local-language KB.

Pipeline. For a regional-language question, we internally translate a copy of the question to English (using a translation-only prompt) solely for KB searching → hybrid BM25 + dense retrieval against the **English** cultural KB already built for PS-1 → retrieved English passages are attached to the **original regional-language question** → the LLM answers in the format BLEnD expects. Thus, the query used for retrieval and the question answered by the LLM are kept separate: English is used for matching against the English KB, while the final question remains in its original regional language. Fig. 4 shows this pipeline.

Comparison conditions (on the 3 low-resource countries only). (A) *No-retrieval baseline* in the regional language (already produced by PS-1, reused here). (B) *English-pivot retrieval* as described above, using the same English KB and hybrid BM25 + dense retrieval pipeline. The regional-language question is translated only for retrieval; the final generation prompt retains the original regional-language question.

Feasibility bound. This is restricted to the 3 low-resource countries (not all 6), keeping the added call volume to roughly the size of one additional PS-1-scale run rather than doubling the whole project.

Honest framing. We do not assume cross-lingual pivoting will improve performance – translation can lose culturally-specific nuance and can suppress retrieval quality. We report condition (B) vs. (A) as a hypothesis test.

D. PS-4: Category-Aware Analysis (Member 2)

Motivation. BLEnD questions span six everyday categories (food, sports, family, education, holidays/celebrations/leisure, work-life). Reporting only overall per-country accuracy can hide that retrieval helps some domains and hurts others.

Analysis, not new inference. Category-aware analysis is computed entirely from the result logs already produced by PS-1, PS-2, and PS-3 – it requires no additional LLM calls, keeping the overall API budget unchanged. For every question

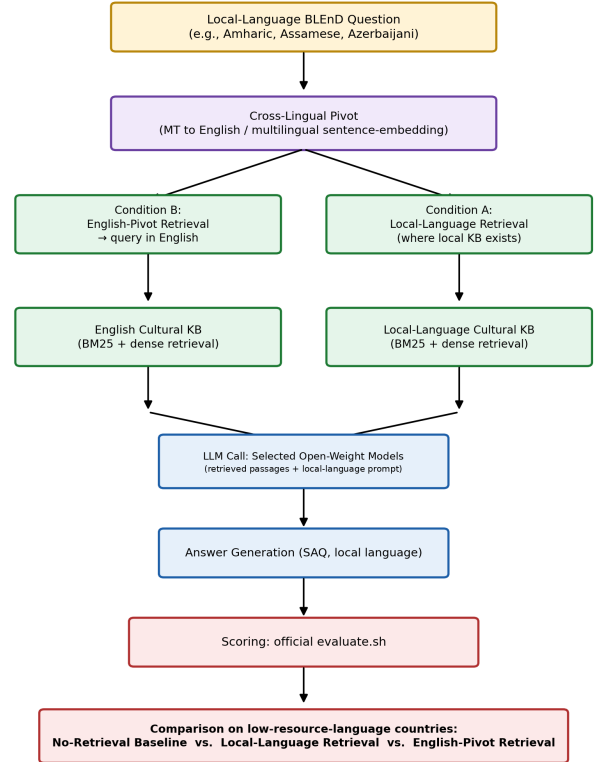


Fig. 4. PS-3 pipeline: a local-language question is optionally pivoted to English before hybrid retrieval, so that low-resource-language questions can draw on the (larger) English cultural knowledge base. English-pivot retrieval is compared against the no-retrieval baseline on the 3 low-resource countries.

we already know its category (from BLEnD’s own metadata), so this is a re-slicing of existing outputs.

Planned tables/plots. (i) Country × Category accuracy (Baseline). (ii) Method × Category accuracy (Baseline vs. RAG vs. Adaptive-RAG, averaged across countries). (iii) Δ -accuracy-by-category heatmap (RAG – Baseline, and Adaptive-RAG – Baseline). (iv) High-resource vs. low-resource category-level comparison, to test whether cultural knowledge gaps are category-dependent rather than uniform.

Questions this answers. Which categories benefit most from RAG; which benefit most from Adaptive-RAG; whether retrieval hurts any category systematically; and whether low-resource cultures show larger category-specific gains (e.g., holidays/celebrations, where locally-sourced KB passages may add the most new information, vs. family, which may already be well covered by an LLM’s parametric knowledge). We report these as empirical findings, not assumed outcomes.

Components built from scratch: knowledge-base scraper/curator, hybrid BM25+dense retrieval module, the retrieval-need decision gate (PS-2), the cross-lingual pivot module (PS-3), prompt-construction and inference wrappers for all five local LLMs, and the ablation/category-analysis scripts. The dataset, question templates (including category labels), and official scoring script (`evaluate.sh`) are reused as released by the original authors.

IV. RESOURCE AND DATASET AVAILABILITY

Dataset: BLEnD (Myung et al., 2024) [1]. Source: <https://github.com/nlee0212/BLEnD> (code + data) and <https://huggingface.co/datasets/nayeon212/BLEnD>. License: CC BY-SA 4.0.

- **Size/structure:** 52.6k question-answer pairs total (15k short-answer, 37.6k multiple-choice) across 16 countries/regions and 13 languages, human-annotated (5 annotators/question). We use the Short-Answer Question (SAQ) subset only, restricted to **6 countries:** United States, Spain, South Korea (high/mid-resource) and Azerbaijan, Assam, Ethiopia (low-resource) – roughly 500 questions/country, i.e. $\sim 3,000$ questions total, evaluated in both English and local language where available.
- **Preprocessing:** No modification to gold data. We reuse the paper’s own prompt templates (`data/prompts/`) and lemmatiser/stemmer-based scoring exactly as released, so our numbers remain directly comparable to the paper’s published baselines.
- **Justification:** This subset spans the full resource spectrum documented in the original paper (its Table 4 resource-availability classes 1–5), which is exactly the axis our RAG intervention is designed to affect, while keeping inference volume small enough to run on the available GPU resources within 2 months.

Knowledge-base construction (for RAG, not part of BLEnD): short, factual cultural-context passages (~ 50 – 100 per country) manually curated/scraped from public Wikipedia and Wikivoyage pages for the 6 target countries – built independently of BLEnD to avoid data contamination.

Resources:

- **Hardware:** A GPU is required for local inference of the selected open-weight LLMs. We use a 32GB GPU for model inference and embedding computation; no fine-tuning is performed. BM25, scoring, and analysis remain lightweight local computation.
- **Software:** Python 3, `rank_bm25` or `elasticsearch-lite` for lexical retrieval, `sentence-transformers` for dense embeddings (open, free, CPU-friendly), the original BLEnD repository’s evaluation code, `pandas/matplotlib` for analysis.
- **Embedders for Semantic Search:** OpenAI `text-embedding-3-small` (as a high-performance commercial alternative) or BAAI/`bge-m3` (multilingual, state-of-the-art open-source).
- **Models used (free, non-proprietary, no paid API):**
 - **Qwen3-8B** – open-weight, locally deployable model under the Apache 2.0 license.
 - **Aya-101** – open-weight, locally deployable model under the Apache 2.0 license.
 - **GPT-OSS-20B** – open-weight, locally deployable model.
 - **Gemma 2** – open-weight, locally deployable model.
 - **Meta Llama 3.1** – open-weight, locally deployable model.

Resource management: We deliberately restrict the project to five free, non-proprietary, open-weight models (Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, Meta Llama 3.1) rather than the paper’s proprietary API models or its larger GPU-heavy roster, since our revised setup is intended to remain reproducible without paid API access. We now require GPU resources for local model inference, but the selected models are sized to fit the available 32GB GPU without fine-tuning. We explicitly report this as a scope limitation (Section V) rather than attempting a partial, resource-constrained reproduction of the full 16-model comparison.

V. LIMITATIONS FOUND

Limitations of the original paper/baseline:

- Zero-shot prompting alone (the paper’s only tested method) leaves a very large accuracy gap between high- and low-resource cultures (up to 67 points), with no mitigation proposed within the paper itself.
- The paper’s exact-match/lemma-based scoring is known (confirmed independently by Team GUIR [7] in the SemEval-2026 follow-up) to *underestimate* true model performance by up to ~ 18 points for morphologically rich languages – we will report this as a caveat when interpreting our own low-resource-language results, rather than attempting to fix the metric ourselves (out of scope for 2 months).
- The paper’s original 16-model evaluation used paid APIs and an 8-GPU cluster, which is not reproducible on a student budget.

Additional scope caveats introduced by the new components (flagged explicitly, per project audit):

- **Adaptive retrieval (PS-2)** uses a heuristic gate, not a learned or fine-tuned classifier – its accuracy at predicting “retrieval will help” is itself an empirical question, not guaranteed to be well-calibrated.
- **Cross-lingual retrieval (PS-3)** depends on machine-translation quality for the pivot step; translation errors on culturally-specific terms could suppress rather than help retrieval quality, and we report this as a possible failure mode rather than assume it away.
- Both PS-2 and PS-3 are restricted in scope (PS-2 reuses PS-1’s retrieval signal; PS-3 covers only the 3 low-resource countries) specifically so that adding them does not multiply the total LLM call budget – see the compact experiment matrix in Table II.

How we address these:

- We directly target the performance-gap limitation with a concrete inference-time intervention (region-aware RAG) rather than only re-measuring the existing gap.
- We do not claim to fix the scoring-metric limitation; we explicitly flag it as an interpretation caveat for any near-miss/false-negative cases we observe in low-resource languages.
- We address the compute/budget limitation by scoping to 3 free, non-proprietary models and 6 countries, chosen to preserve the resource-level diversity that matters for

our research question, rather than attempting a diluted, low-quality version of the full 16-model sweep.

What differentiates our work: Unlike the best-performing SemEval-2026 RAG system (Team king001), whose reported gains were confounded by likely contamination from BLEnD’s public 2024 release, our knowledge base is built entirely from independent, non-BLEnD sources. Our central contribution is not a single leaderboard number but a **per-country ablation** that tests, and reports honestly, *where* retrieval augmentation helps (expected: low-resource cultures such as Ethiopia, Assam, Azerbaijan) and where it may not, or could even hurt (expected: already-high-performing cultures such as the US), following the nuanced finding reported by Team CultRAG on this same benchmark.

Expected improvement (hypothesis, to be validated empirically – not a promised result): The one rigorously-measured RAG-vs-baseline comparison available to us from the SemEval-2026 follow-up (Team king001 vs. a plain zero-shot baseline, on held-out, contamination-checked data) showed an *effectively zero* net gain (59.5 vs. 59.5 average) once contamination was controlled for. We take this as our realistic reference point rather than an optimistic one. Our working hypothesis is that selected open-weight models may have more headroom on low-resource cultures than the frontier model king001 used, which is why RAG (and, more targetedly, Adaptive-RAG) *might* produce a small positive, low-resource-skewed delta – but we explicitly do not promise a specific percentage-point gain for any country, model, or condition. Success is defined as *any* consistent, honestly-measured delta pattern (positive, null, or negative) reported per country/model/condition, not as hitting a target number. This also applies to PS-2 (Adaptive-RAG is not assumed to beat Always-RAG) and PS-3 (English-pivot retrieval is not assumed to improve over the no-retrieval baseline).

VI. EXPECTED OUTCOMES

Performance metrics targeted:

- SAQ accuracy (official BLEnD lemma/stem-matching scorer) for Baseline vs. Always-RAG vs. Adaptive-RAG, for each of 6 countries $\times$ 5 models (90 result cells total across three conditions).
- SAQ accuracy for no-retrieval baseline vs. English-pivot cross-lingual retrieval, for the 3 low-resource countries $\times$ 5 models.
- Aggregate “high-resource vs. low-resource” accuracy gap, across all conditions, as the primary summary statistic of improvement.
- Category-level accuracy (Country $\times$ Category, Method $\times$ Category) computed from the above, without additional inference calls.
- Fraction of questions on which the Adaptive-RAG gate chooses to retrieve, by country resource-level – a diagnostic, not an accuracy metric.
- **No fixed numeric target** is set for any delta (see Section V); success is a complete, honestly-reported set of deltas across all conditions and categories.

Final deliverables:

TABLE I

PLANNED RESULT LAYOUT (SAQ ACCURACY %). ONLY THE BASELINE COLUMN FOR US/SPAIN/ETHIOPIA IS A REAL, PAPER-SOURCED FIGURE (MYUNG ET AL. 2024, AVERAGED ACROSS THEIR ORIGINAL 16 MODELS, LOCAL LANGUAGE); ALL OTHER CELLS ARE **TBD**, TO BE FILLED WITH OUR OWN MEASURED QWEN3-8B/AYA-101/GPT-OSS-20B/GEMMA 2/META LLAMA 3.1 NUMBERS ONCE WEEK-1/WEEK-4 RUNS COMPLETE. NO RAG/ADAPTIVE-RAG/CROSS-LINGUAL DELTAS ARE PRE-SPECIFIED AS TARGETS – SEE SECTION V FOR WHY.

Country	Baseline*	+RAG	+Adapt.	+X-Lingual
United States (high)	~79*	TBD	TBD	–
Spain (mid-high)	~69*	TBD	TBD	–
South Korea (mid)	TBD	TBD	TBD	–
Azerbaijan (low)	TBD	TBD	TBD	TBD
Assam (low)	TBD	TBD	TBD	TBD
Ethiopia (low)	~12*	TBD	TBD	TBD

TABLE II

COMPACT EXPERIMENT MATRIX. CROSS-LINGUAL RETRIEVAL (PS-3) IS SCOPED TO THE 3 LOW-RESOURCE COUNTRIES ONLY, SO IT DOES NOT MULTIPLY THE FULL 6-COUNTRY MATRIX; CATEGORY-AWARE ANALYSIS (PS-4) IS A RE-SLICING OF THESE SAME RESULT CELLS, ADDING NO FURTHER LLM CALLS.

Condition	Countries	Models	Owner (PS)
Baseline (direct+persona)	6	3	PS-1 (M1)
Always-RAG	6	3	PS-1 (M1)
Adaptive-RAG	6	3	PS-2 (M2)
Cross-Lingual (Eng.-pivot)	3	3	PS-3 (M1)
Cross-Lingual (Eng.-pivot)	3	3	PS-3 (M1)
Category-aware analysis	re-slices rows above		PS-4 (M2)

- Full reproducible codebase: knowledge-base builder, hybrid retrieval module, the adaptive retrieval-need gate, the cross-lingual pivot module, baseline/RAG/Adaptive-RAG/cross-lingual inference pipelines for Qwen3-8B, Aya-101, GPT-OSS-20B, Gemma 2, and Meta Llama 3.1, and analysis scripts.
- Results report with per-country/per-model/per-condition tables (Table I, Table II) and comparison against the original paper’s published baseline numbers.
- Category-aware tables and plots (Country $\times$ Category, Method $\times$ Category, Δ -by-category) as described in Section III-D.
- Ablation analysis and error analysis (qualitative examples of where RAG/Adaptive-RAG/cross-lingual retrieval help or hurt), including discussion of the SAQ scoring-metric limitation noted in Section V.
- Final written report / paper documenting the pipeline, results, and honest discussion of limitations (including any null results).

Stretch goals (only if ahead of schedule): MCQ cross-locale majority voting (GUIR’s technique [7]); extending cross-lingual retrieval experiments if additional local-language retrieval becomes feasible within the main timeline. Stretch goals do not replace or delay PS-1–PS-4.

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